Xu Jian

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Curriculum Vitae

Education

2013.09–2017.07 **Bachelor**, *Communication University of China*, Beijing, China, *Business Administration*

Second Prize in the National College Student Mathematics Competition

2018.09–2020.07 **Master**, *South China University of Technology*, Guangzhou, China, *Basic Mathematics*

First Class Academic Scholarship from South China University of Technology, Advisor: Associate Professor Bingsheng Lin

2020.09–2025.12 **PhD**, *South China University of Technology*, Guangzhou, China, *Computational Mathematics & Electronic Information*

He has authored five first-author papers in top-tier conferences including ICML Oral and ICLR, and three first-author papers published in prestigious journals, including IEEE Transactions on Neural Networks and Learning Systems (TNNLS) and Knowledge- based Systems (KBS).

Internship Experience

2025.07 – 2026.01 RIKEN AIP, Tokyo, Japan Research Intern

Work Experience

2026.02–Present **Postdoc Researcher**, *RIKEN Interdisciplinary Theoretical and Mathematical Sciences Program (iTHEMS) & RIKEN Center for Advanced Intelligence Project (AIP)*, Tokyo, Japan

Conduct research on probabilistic machine learning and stochastic dynamical systems. Focus on generative modeling, stochastic differential equations, and interdisciplinary applications in physics-inspired AI.

Research Focus

2020-Present Probabilistic Machine Learning, Stochastic Process Methods, Generative Models and Applications, Differential Equations and Dynamical Systems, Quantum Computing, etc.

Notes

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Google Scholar https://scholar.google.com/citations?user=DublkSoAAAAJ&hl=zh-CN

Research Achievements

Published Papers

1 **Xu, J.**, & Zeng, D. (2024, March). **Sparse Variational Student-t Processes**. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 38, No. 14, pp. 16156-16163).

2 **Xu, J.**, Zeng, D.& Paisley, J., (2024). **Sparse Inducing Points in Deep Gaussian Processes: Enhancing Modeling with Denoising Diffusion Variational Inference**. In Proceedings of the 41st International Conference on Machine Learning, PMLR 235:55490-55500, 2024. (**Oral Top 1% in submitted papers**)

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3 **Xu, J.**, Zeng, D. (2025) & Paisley, J.,. **Diffusion Bridge Variational Inference for Deep Gaussion Processes** Accepted to ICLR 2026. https://openreview.net/forum?id=zyRmy0Ch9a

4 **Xu, J.**, Du, S., Yang, J., Ma, Q., & Zeng, D. (2024). **Neural Operator Variational Inference Based on Regularized Stein Discrepancy for Deep Gaussian Processes**. In IEEE Transactions on Neural Networks and Learning Systems, doi: 10.1109/TNNLS.2024.3406635.

5 **Xu, J.**, Du, S., Yang, J., Ding, X., Zeng, D., & Paisley, J. (2025, April). **Bayesian Gaussian Process ODEs via Double Normalizing Flows**. In International Conference on Artificial Intelligence and Statistics (pp. 235-243). PMLR.

6 **Xu, J.**, Zeng, D.& Paisley, J., (2024). **Fully Bayesian Differential Gaussian Processes through Stochastic Differential Equations**. Knowledge-Based Systems, Volume 314, 2025, 113187, ISSN 0950-7051, https://doi.org/10.1016/j.knosys.2025.113187.

7 **Xu, J.**, Du, S., Yang, J., & Zeng, D. (2023). **Variational Learning of Gaussian Process Latent Variable Models through Stochastic Gradient Annealed Importance Sampling**. Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence, PMLR 286:4663-4680, 2025.

8 **Xu, J.**, Zeng, D. (2024) & Paisley, J.,. **Sparse Variational Student-t Processes for Heavy-tailed Modeling**. In IEEE Transactions on Neural Networks and Learning Systems, doi: 10.1109/TNNLS.2026.3673350.

9 Du, S., Luo, Y., Chen, W., **Xu, J.**, & Zeng, D. (2022). **To-flow: Efficient continuous normalizing flows with temporal optimization adjoint with moving speed**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 12570–12580).

10 Lin, Z., Chen, W., **Xu, J.**, Zeng, D., & Chen, M. (2024). **ReciprocalLA-LLIE: Low-light image enhancement with luminance-aware reciprocal diffusion process**. *Neurocomputing*, 131438.

11 Lin, B., **Xu, J.**, & Heng, T. (2019). **Induced entanglement entropy of harmonic oscillators in non-commutative phase space**. *Modern Physics Letters A*, 34(33), 1950269.

12 Fu, H., **Xu, J.** (2026). **Multi-Scale Convolution with Optimal Transport Attention Effect on Multivariate Time Series**. *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*.

13 Li, W., Xu, Z., **Xu, J.**, Li, L., & Bai, L. (2025). **Binomial jump-amplitude modeling in SDEs: A regularized stepwise estimation with computational diagnostics**. *AIMS Mathematics*, 10(11), 26994–27015.

14 Li, W., Jiang, Y., **Xu, J.**, Wang, L., & Bai, L. (2025). **Enhanced parameter estimation for stable SDEs with jumps via quadratic variation**. *Journal of Statistical Computation and Simulation*, 95(13), 2932–2948.

15 Li, W., Zhang, L., **Xu, J.**, Li, L., & Bai, L. (2025). **Jump amplitude inference in SDEs with cosine kernel**. *Results in Applied Mathematics*, 26, 100596.

Submitted Papers

16 **Xu, J.**, Zeng, D. (2024) & Paisley, J.,. **Flexible Bayesian Last Layer Models Using Implicit Priors and Diffusion Posterior Sampling**. Submitted to EAAI

17 **Xu, J.**, Zeng, D. (2024) & Paisley, J.,. **Implicit Variational Reject Sampling**. Submitted to UAI 2026

18 **Xu, J.**, Zeng, D. (2025) & Paisley, J.,. **Denoising Diffusion Variational Inference for Bayesian Deep Models: A Unified Framework for Sparse Inducing Variable Optimization** Submitted to TPAMI

19 **Xu, J.**, Zeng, D. (2025), Zhao, Q. & Paisley, J.,. **Neural Bridge Process** Submitted to Neurips 2026

20 **Xu, J.**, Zeng, D. (2026) Paisley, J.,.& , Zhao, Q. **Consist-Retinex: One-Step Noise-Emphasized Consistency Training Accelerates High-Quality Retinex Enhancement** Submitted to Neurips 2026

21 **Xu, J.**, Zeng, D., Paisley, J., & Zhao, Q. (2026). **Stochastic Schrödinger Diffusion Models for Pure-State Ensemble Generation**. Submitted to NeurIPS 2026.

22 **Xu, J.**, Li, C., Zeng, D., Paisley, J., & Zhao, Q. (2026). **Intrinsic Flow Matching on Quantum Pure-State Manifolds with Phase-Aligned Transport**. Submitted to NeurIPS 2026.

23 **Xu, J.**, Li, C., Zeng, D., Paisley, J., & Zhao, Q. (2026). **AQKA: Active Quantum Kernel Acquisition Under a Shot Budget**. arXiv:2605.14672.

Project Experience

Dec 2023 – Feb 2024 Assisted in the preparation of a National Natural Science Foundation of China (NSFC) proposal titled "*Intelligent Emergence Mechanisms of Generative Models*" . Contributed to theoretical framework design and technical route formulation.

May 2024 – Present Paper Lecturer, Turing Academy (Subsidiary of Dark Horse Technology Group). Provide mentorship on academic writing, research methodology, and publication strategies. Guided multiple students in preparing and submitting papers to top-tier conferences and SCI-indexed journals.

Feb 2026 – Present Principal Investigator, RIKEN Quantum Postdoctoral Research Project. Led research on quantum-inspired probabilistic generative models and stochastic dynamical systems, exploring theoretical foundations and interdisciplinary applications in quantum computing and AI.